Code:

```
#include <stdio.h>
#include <stdlib.h>

// Structure to represent a block
struct Block {
   char data[256]; // Adjust size if needed
   struct Block* next;
};

int main()
{
   struct Block* firstBlock = NULL; // First block
   struct Block* lastBlock = NULL;  // Last block

   int blockCount = 0;

   int blockNumber;
   char data[256];
   char choice;

   printf("Linked Allocation Simulation\n");

   while (1)
   {
      printf("Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: ");
      scanf(" %c", &choice);

      if (choice == 'Q' || choice == 'q')
      {
         break;
      }

      if (choice == 'W' || choice == 'w')
      {
         printf("Enter data for the block: ");
         scanf(" %[^\n]", data);

         // Create new block
         struct Block* newBlock = (struct Block*)malloc(sizeof(struct Block));

         // Copy data
         for (int i = 0; i < 256; i++)
         {
            newBlock->data[i] = data[i];
         }

         newBlock->next = NULL;

         if (blockCount == 0)
         {
            firstBlock = newBlock;
            lastBlock = newBlock;
         }
         else
         {
            lastBlock->next = newBlock;
            lastBlock = newBlock;
         }

         blockCount++;
      }
      else if (choice == 'R' || choice == 'r')
      {
         printf("Enter the block number to read (1-%d): ", blockCount);
         scanf("%d", &blockNumber);

         if (blockNumber < 1 || blockNumber > blockCount)
         {
            printf("Invalid block number. The valid range is 1-%d.\n", blockCount);
         }
         else
         {
            struct Block* currentBlock = firstBlock;
            for (int i = 1; i < blockNumber; i++)
            {
               currentBlock = currentBlock->next;
            }

            printf("Block %d Data: %s\n", blockNumber, currentBlock->data);
         }
      }
   }

   // Free memory
   struct Block* currentBlock = firstBlock;
   while (currentBlock != NULL)
   {
      struct Block* nextBlock = currentBlock->next;
      free(currentBlock);
      currentBlock = nextBlock;
   }

   return 0;
}
```

Output:

C:\Users\Dell\Downloads\Unt

```
Linked Allocation Simulation
Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: W
Enter data for the block: HI HOW ARE YOU
Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: W
Enter data for the block: THIS IS ASWATHA
Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: R
Enter the block number to read (1-2): 1
Block 1 Data: HI HOW ARE YOU
Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: R
Enter the block number to read (1-2): 2
Block 2 Data: THIS IS ASWATHA
Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: Q

Process returned 0 (0x0)   execution time : 49.026 s
Press any key to continue.
```